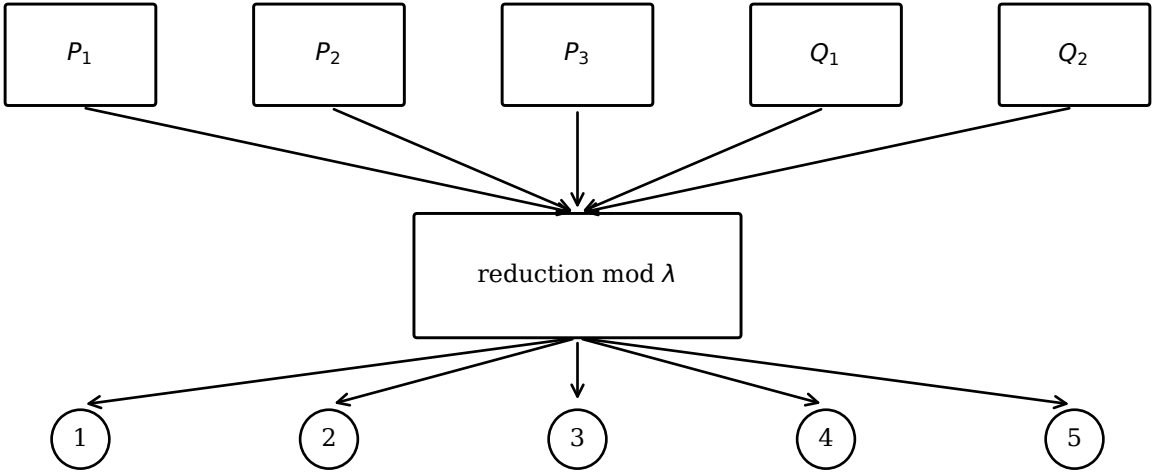A separating prime preserves every distinction in the finite polynomial alphabet